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AI as Adjudicator: Evaluation infrastructure for accountable AI subjective grading in education

Presenter

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Collaborators

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This entry presents the next step of a programme whose foundation — "Beyond Teacher Agreement: Reference-Label Validity in AI-Assisted Subjective Assessment" (the App-Phase validation) — is a working paper now being prepared for submission to a peer-reviewed journal. That work established adjudication with a human adjudicator; here the adjudicator becomes a model-invariant AI panel that runs on every case. Foundation results are cited; full results are in that working paper (to be communicated).

Summary

As AI enters real assessment, the question is not "which grader is right?" but "who checks the evaluator?" — and the answer has moved through three stages. Stage 1 — teacher as ground truth: scoring the AI by its agreement with the teacher measures resemblance to a label, not the label's validity (teacher marks carry rater severity, fatigue, rubric ambiguity, missed evidence), so a disagreement is resolved by assumption, not evidence. Stage 2 — human as adjudicator (our foundation, under EM:IP review): a human judges each disputed mark against the policy-derived scheme, not the teacher — validated on a government-school study (ICC 0.930) — but a human is fallible, non-reproducible, and affordable only on disputes. Stage 3 — AI as adjudicator, with evidence and an audit trail (this work): a model-invariant panel + a deterministic cascade resolve AI-vs-teacher grading reproducibly and by conformance to policy, anchored to a verifiable evidence span. Cheap and automated, it runs on every case — surfacing the shared blind spot: where teacher and AI agree and are both wrong, invisible to any disagreements-only method.

Methodology

Both adjudicators run on one evaluation infrastructure — an API combining OCR (handwriting), an LLM, human-in-the-loop review, and a full audit trail, marking each answer against the board policy and scheme in force, every score traceable to its rubric cell (the Evaluate component of a wider suite). The Stage-2 (human) adjudicator followed a documented, provenance-masked protocol, coding each disagreement with a rationale and a mismatch-driver under five governance functions (contestability, selective oversight, accountability, safeguards, preserved human authority). The Stage-3 adjudicator (this work) replaces

the human with a panel of four independent LLM families marking one value-point at a time (temperature 0, three draws), feeding a fully deterministic four-step cascade — admissibility → defensibility → correct-mark recovery → evidence — with no model in the loop at decision time. Human authority is preserved: the cascade auto-resolves what it is confident about (every verdict reviewable and overridable) and refers the rest to a human, evidence attached.

Figure 1. The four-step deterministic adjudication cascade and its resolution funnel.



Novelty

Three novel contributions (to our knowledge, in combination):

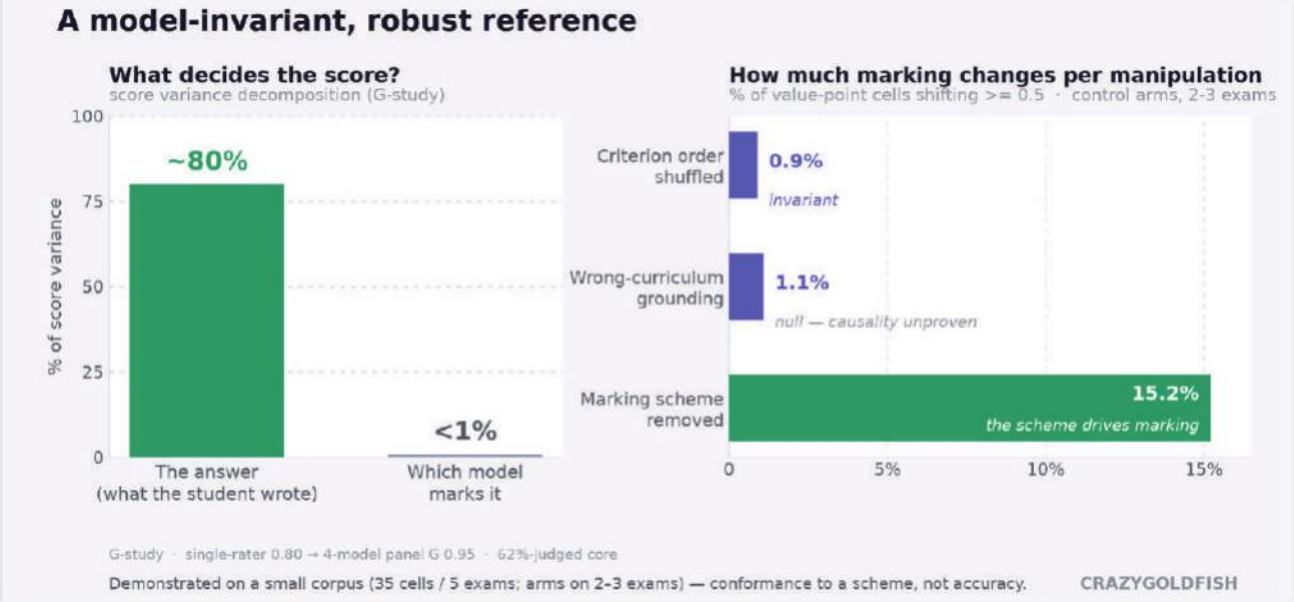
1. **The adjudicator moves from human to machine, and from disputes to every case.** Being repeatable, scalable, and cheap, the panel adjudicates agreements too — surfacing the shared blind spot: both markers award full marks for two of eight required journal entries a student never attempted (defensible mark 3, not 4). An entry is present or not — an objective error a "teacher 4, AI 4" review never looks for. The panel also recovers marks both deny (both gave 0 where the student earned credit) — protecting the student, not only auditing the machine.

2. **A two-level policy grounding qualifies the panel as an arbiter.** A neutral arbiter must be independent of the parties and anchored to an authority both follow. (Level 1) the scheme is derived from board policy — model-independent, the same criteria for every grader, refusing to invent structure where policy is silent (holistic items → human review). (Level 2) that scheme is marked model-invariantly — <1% of a score depends on which model; ~80% on the answer (G = 0.95) — every credit carrying a verifiable span. Neither level alone suffices; together they make the panel a policy-anchored, grader-invariant

reference, not a fifth opinion. (We claim policy-derivation, model-invariance, and auditability — not that the policy content is causally proven.)

3. **Deterministic adjudication — reproducible where a human cannot be.** A human adjudicator is fallible and non-reproducible; the cascade yields an identical, reconstructable verdict every run, resolving 82% of cases deterministically. A self-preference guard confirms the panel does not favour its own lineage — removing the AI's own model family does not collapse its verdicts. (How the invariant scheme and marker are built is confidential; the adjudication methodology is disclosed in full.)

Figure 2. Model-invariance and robustness — cross-family agreement, generalisability, and the self-preference guard.



Validation

Foundation (App-Phase validation, cited): a bounded 23-copy government-school study reported strong copy-level agreement (ICC(A,1) = 0.930); full results and limitations are in the manuscript under EM:IP review. This work (the AI panel): cross-family agreement (Gwet AC2) 0.768 ± 0.162 over 35 cells / 5 exams (per-exam 0.699–0.816); generalisability G = 0.95 (<1% from the model, ~80% from the answer; 62%-judged core), reproduced three independent ways to ≤0.01; ~82% of awarded credits span-verified. Run on all 56 question-cases, the cascade resolves 46 (82%) deterministically — teacher-better 5 · AI-better 9 · both-defensible 24 · neither-defensible 8 — and refers 10 (18%) to a human with evidence; that 82% reproduces every run, a human's verdicts do not. Honest scope: demonstrated mechanisms on a small corpus, not population rates; conformance to a scheme, not accuracy against truth; the two adjudicators are not like-for-like (the human addressed only disputes; the panel, all cases). Larger-N validation and a human-expert accuracy check are the defined next steps. Maturity: the Evaluate platform is in

production (≈TRL 8; MeitY Case Study 8); the adjudication method here is an earlier-stage research demonstration.

Future Scope

Live in production with paying customers across K-12 government (MeitY reference), private school chains (multi-board, embeddable LMS UI), and NGO programmes (Hindi-medium; internal reliability α = 0.808), with a HigherEd multi-domain pilot pending. Research agenda: larger-corpus rates and multi-adjudicator reliability. Beyond Evaluate, the roadmap extends to Personalise, Generate, Observe, and Engage on the same adjudicated foundation.

Market	~250M+ school learners; 8M+ teachers; 15L+ schools in India; NEP 2020 + NCF 2023 mandate competency-based assessment
Business model	Per-answer, pay-as-you-go — no platform fee; the scheme is authored once per exam and amortised across every student, so cost scales with answers evaluated. "We monetise the evaluation layer, not the data."
Positioning	Embedded, policy-aware grading with a documented, auditable adjudication methodology — an evaluation-governance layer, not a standalone grader

Speaker Bio



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Rahul Khandelwal is the Founder & CEO of CrazyGoldFish, where he is building evaluation infrastructure for education — the memory layer that turns subjective assessment into validated, reusable learning signal. Before CrazyGoldFish, he led large-scale personalised learning (Pratham, 2.3M children), adaptive remediation (UnlearnKids), and product and technology as VP at CollegeDekho. His current research reframes how AI grading is judged — moving the reference for a disputed mark from a single human adjudicator to a model-invariant, reproducible AI panel, so evaluation can be held to account at national scale. He is driven by one conviction: in education, AI must be measured against defensible evidence, not convenient assumptions, and final judgment must remain human.